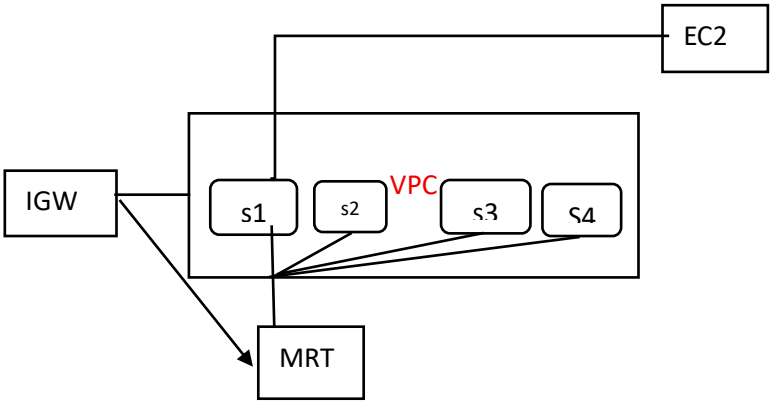
Name: shravani Akude

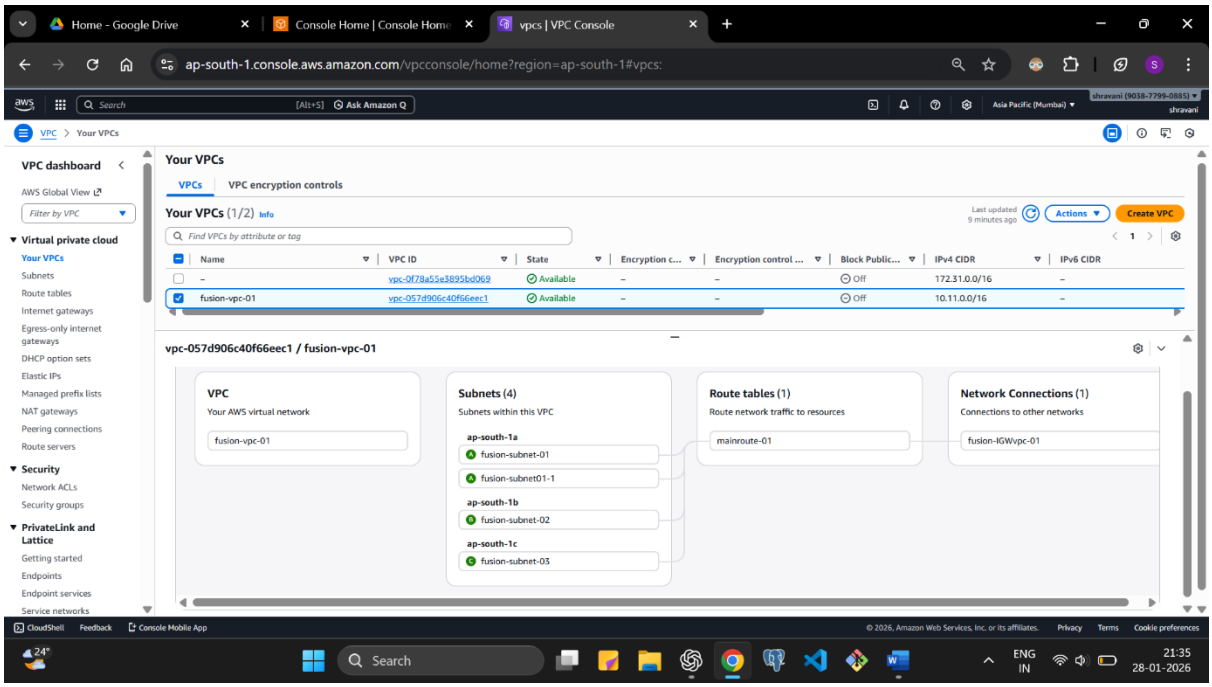
Batch no – 074

Topic – VPC creation with both package installation

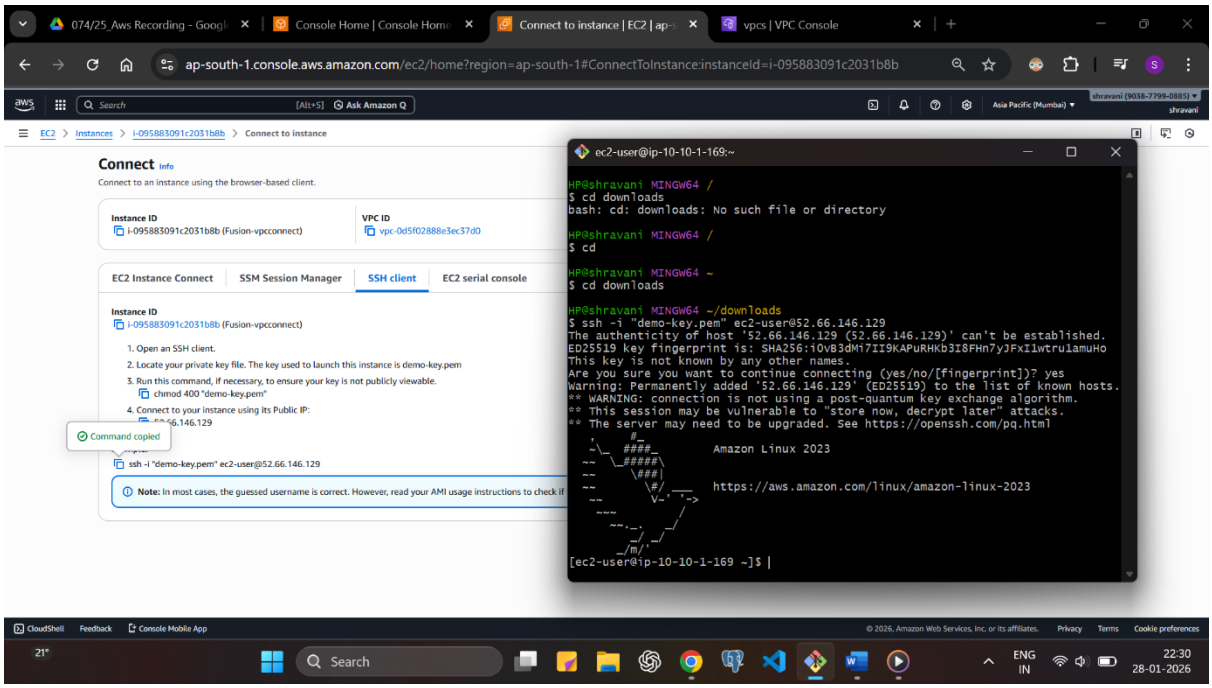


- **Steps to ec2 connection and package installation**

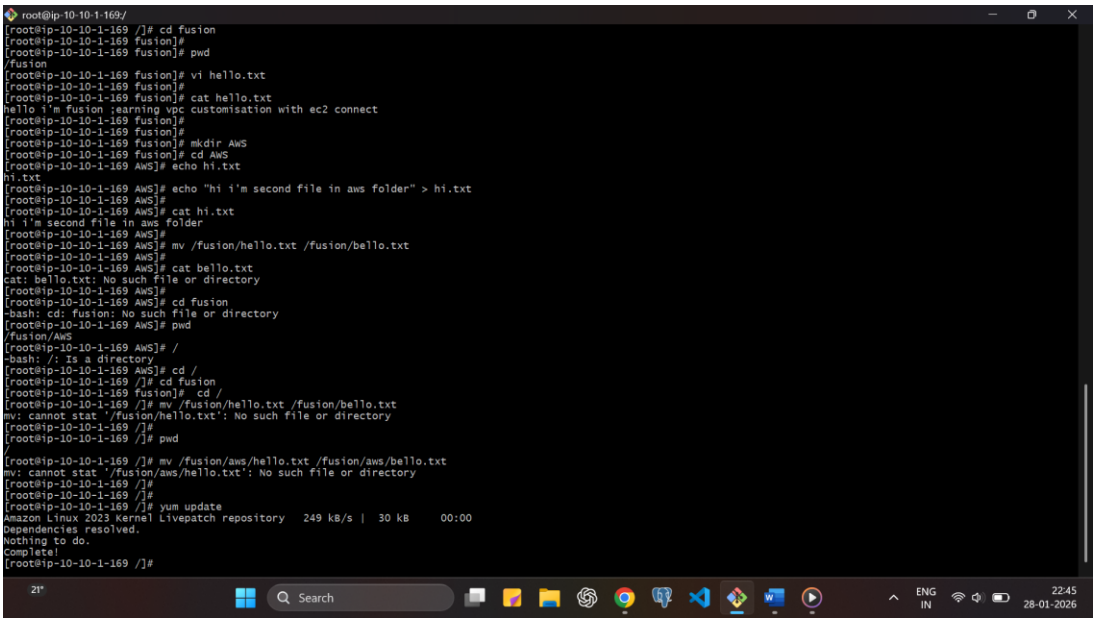
1. First create VPC subnets, IGW and connect them after that go to EC2 launch it new tab



2. Give instance name and click in edit vpc select our created VPC and enable it launch ec2, Connect ec2 to gitbash.



3. go to downloads after that to update yum command is used



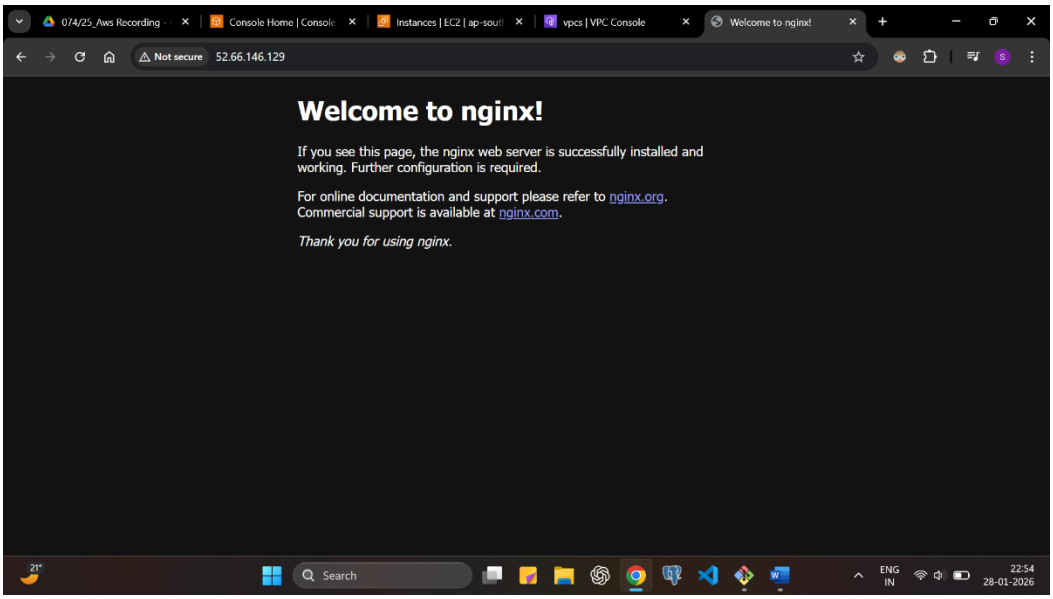
4. to install nginx yum install nginx is used then check the status to check systemctl status nginx

```
root@ip-10-10-1-169:/
Installed size: 3.7 M
Is this ok [y/N]: y
Downloading Packages:
(1/7): libunwind-1.4.0-5.amzn2023.0.3.x86_64.rp 1.7 MB/s |  66 kB     00:00
(2/7): generic-logos-httpd-18.0.0-12.amzn2023.0 463 kB/s |  19 kB     00:00
(3/7): gperftools-libs-2.9.1-1.amzn2023.0.3.x86 6.5 MB/s | 308 kB     00:00
(4/7): nginx-1.28.1-1.amzn2023.0.1.x86_64.rpm   1.3 MB/s |  33 kB     00:00
(5/7): nginx-core-1.28.1-1.amzn2023.0.1.x86_64.  22 MB/s | 687 kB     00:00
(6/7): nginx-filesystem-1.28.1-1.amzn2023.0.1.n 339 kB/s | 9.6 kB     00:00
(7/7): nginx-mimetypes-2.1.49-3.amzn2023.0.3.no 857 kB/s |  21 kB     00:00
--------------------------------------------------------------------------------
Total                                           8.7 MB/s | 1.1 MB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                        1/1
  Running scriptlet: nginx-filesystem-1:1.28.1-1.amzn2023.0.1.noarch        1/7
  Installing       : nginx-filesystem-1:1.28.1-1.amzn2023.0.1.noarch        1/7
  Installing       : nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch           2/7
  Installing       : libunwind-1.4.0-5.amzn2023.0.3.x86_64                  3/7
  Installing       : gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64            4/7
  Installing       : nginx-core-1:1.28.1-1.amzn2023.0.1.x86_64              5/7
  Installing       : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch      6/7
  Installing       : nginx-1:1.28.1-1.amzn2023.0.1.x86_64                   7/7
  Running scriptlet: nginx-1:1.28.1-1.amzn2023.0.1.x86_64                   7/7
  Verifying        : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch      1/7
  Verifying        : gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64            2/7
  Verifying        : libunwind-1.4.0-5.amzn2023.0.3.x86_64                  3/7
  Verifying        : nginx-1:1.28.1-1.amzn2023.0.1.x86_64                   4/7
  Verifying        : nginx-core-1:1.28.1-1.amzn2023.0.1.x86_64              5/7
  Verifying        : nginx-filesystem-1:1.28.1-1.amzn2023.0.1.noarch        6/7
  Verifying        : nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch           7/7

Installed:
  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
  gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64
  libunwind-1.4.0-5.amzn2023.0.3.x86_64
  nginx-1:1.28.1-1.amzn2023.0.1.x86_64
  nginx-core-1:1.28.1-1.amzn2023.0.1.x86_64
  nginx-filesystem-1:1.28.1-1.amzn2023.0.1.noarch
  nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch

Complete!
[root@ip-10-10-1-169 /]#
[root@ip-10-10-1-169 /]#
[root@ip-10-10-1-169 /]#
```

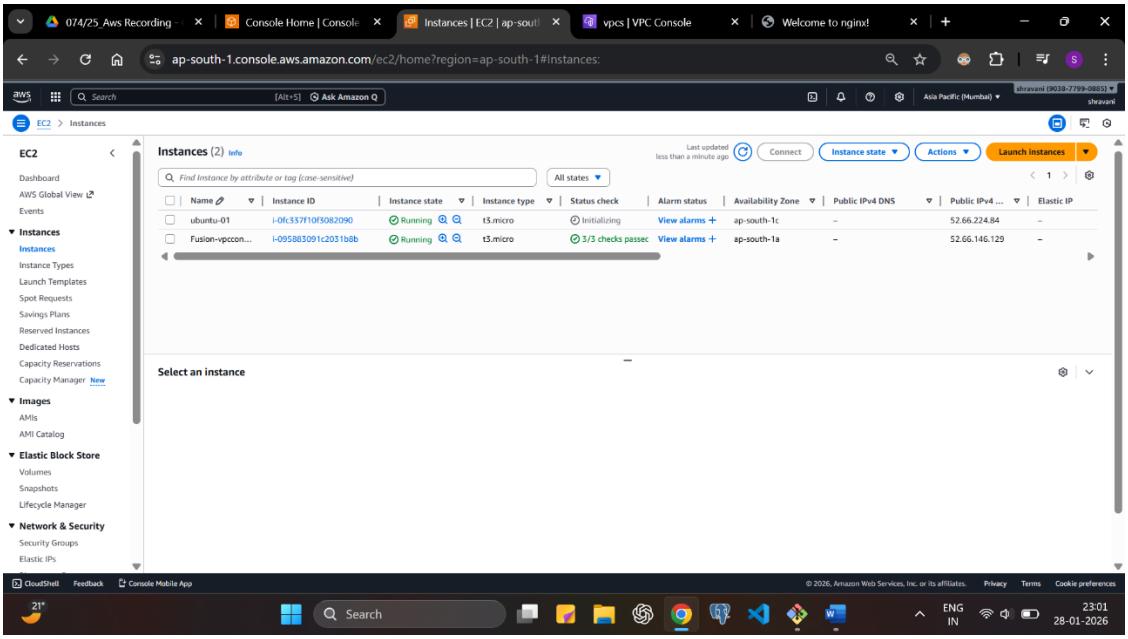
```
root@ip-10-10-1-169:/
  Verifying        : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch      1/7
  Verifying        : gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64            2/7
  Verifying        : libunwind-1.4.0-5.amzn2023.0.3.x86_64                  3/7
  Verifying        : nginx-1:1.28.1-1.amzn2023.0.1.x86_64                   4/7
  Verifying        : nginx-core-1:1.28.1-1.amzn2023.0.1.x86_64              5/7
  Verifying        : nginx-filesystem-1:1.28.1-1.amzn2023.0.1.noarch        6/7
  Verifying        : nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch           7/7

Installed:
  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
  gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64
  libunwind-1.4.0-5.amzn2023.0.3.x86_64
  nginx-1:1.28.1-1.amzn2023.0.1.x86_64
  nginx-core-1:1.28.1-1.amzn2023.0.1.x86_64
  nginx-filesystem-1:1.28.1-1.amzn2023.0.1.noarch
  nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch

Complete!
[root@ip-10-10-1-169 /]#
[root@ip-10-10-1-169 /]#
[root@ip-10-10-1-169 /]# systemctl status nginx
○ nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
     Active: inactive (dead)
[root@ip-10-10-1-169 /]# systemctl start nginx
[root@ip-10-10-1-169 /]#
[root@ip-10-10-1-169 /]# systemctl start nginx
[root@ip-10-10-1-169 /]# systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
     Active: active (running) since Wed 2026-01-28 17:18:31 UTC; 43s ago
    Process: 25770 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
    Process: 25771 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
    Process: 25772 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
   Main PID: 25773 (nginx)
      Tasks: 3 (limit: 1067)
     Memory: 3.2M
        CPU: 55ms
     CGroup: /system.slice/nginx.service
             ├─25773 "nginx: master process /usr/sbin/nginx"
             ├─25774 "nginx: worker process"
             └─25775 "nginx: worker process"

Jan 28 17:18:31 ip-10-10-1-169.ap-south-1.compute.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Jan 28 17:18:31 ip-10-10-1-169.ap-south-1.compute.internal nginx[25771]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Jan 28 17:18:31 ip-10-10-1-169.ap-south-1.compute.internal nginx[25771]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Jan 28 17:18:31 ip-10-10-1-169.ap-south-1.compute.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
[root@ip-10-10-1-169 /]#
```

```
root@ip-10-10-1-169:/
  Verifying        : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch      1/7
  Verifying        : gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64            2/7
  Verifying        : libunwind-1.4.0-5.amzn2023.0.3.x86_64                  3/7
  Verifying        : nginx-1:1.28.1-1.amzn2023.0.1.x86_64                   4/7
  Verifying        : nginx-core-1:1.28.1-1.amzn2023.0.1.x86_64              5/7
  Verifying        : nginx-filesystem-1:1.28.1-1.amzn2023.0.1.noarch        6/7
  Verifying        : nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch           7/7

Installed:
  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
  gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64
  libunwind-1.4.0-5.amzn2023.0.3.x86_64
  nginx-1:1.28.1-1.amzn2023.0.1.x86_64
  nginx-core-1:1.28.1-1.amzn2023.0.1.x86_64
  nginx-filesystem-1:1.28.1-1.amzn2023.0.1.noarch
  nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch

Complete!
[root@ip-10-10-1-169 /]#
[root@ip-10-10-1-169 /]#
[root@ip-10-10-1-169 /]# systemctl status nginx
○ nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
     Active: inactive (dead)
[root@ip-10-10-1-169 /]# systemctl start nginx
[root@ip-10-10-1-169 /]#
[root@ip-10-10-1-169 /]# systemctl start nginx
[root@ip-10-10-1-169 /]# systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
     Active: active (running) since Wed 2026-01-28 17:18:31 UTC; 43s ago
    Process: 25770 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
    Process: 25771 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
    Process: 25772 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
   Main PID: 25773 (nginx)
      Tasks: 3 (limit: 1067)
     Memory: 3.2M
        CPU: 55ms
     CGroup: /system.slice/nginx.service
             ├─25773 "nginx: master process /usr/sbin/nginx"
             ├─25774 "nginx: worker process"
             └─25775 "nginx: worker process"

Jan 28 17:18:31 ip-10-10-1-169.ap-south-1.compute.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Jan 28 17:18:31 ip-10-10-1-169.ap-south-1.compute.internal nginx[25771]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Jan 28 17:18:31 ip-10-10-1-169.ap-south-1.compute.internal nginx[25771]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Jan 28 17:18:31 ip-10-10-1-169.ap-south-1.compute.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
[root@ip-10-10-1-169 /]#
```

5. To start systemctl start nginx is used then copy instance public ip and copy it to web browser



6. To connect it to Ubuntu Launch instance in that click in ubuntu key pair, edit vpc enable it

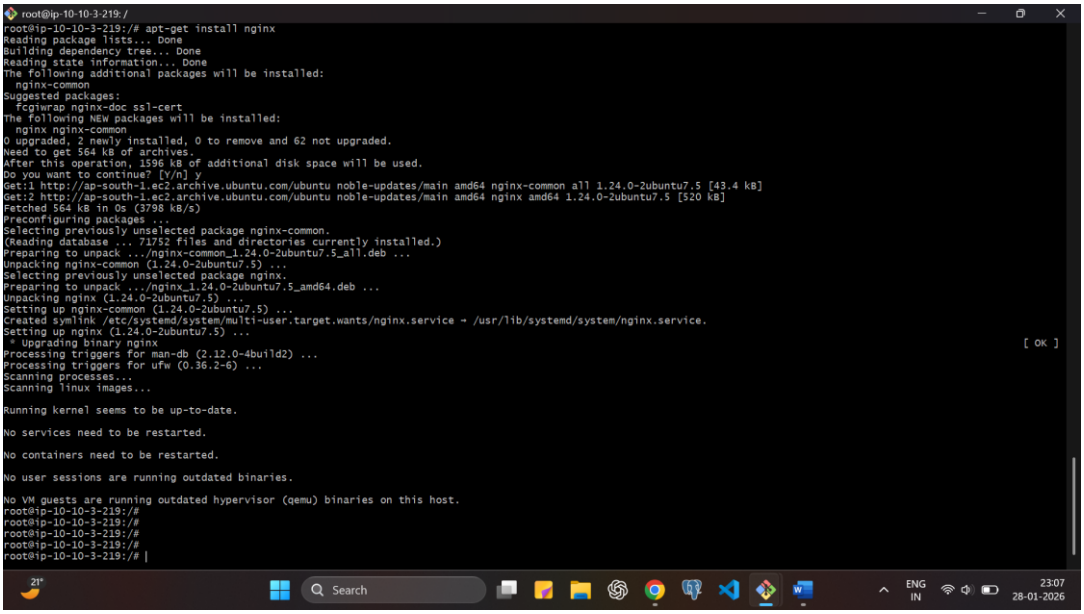


7. open new gitbash copy paste the connect link and it will start after that to install nginx use apt-get install nginx command and to check status use systemctl status nginx

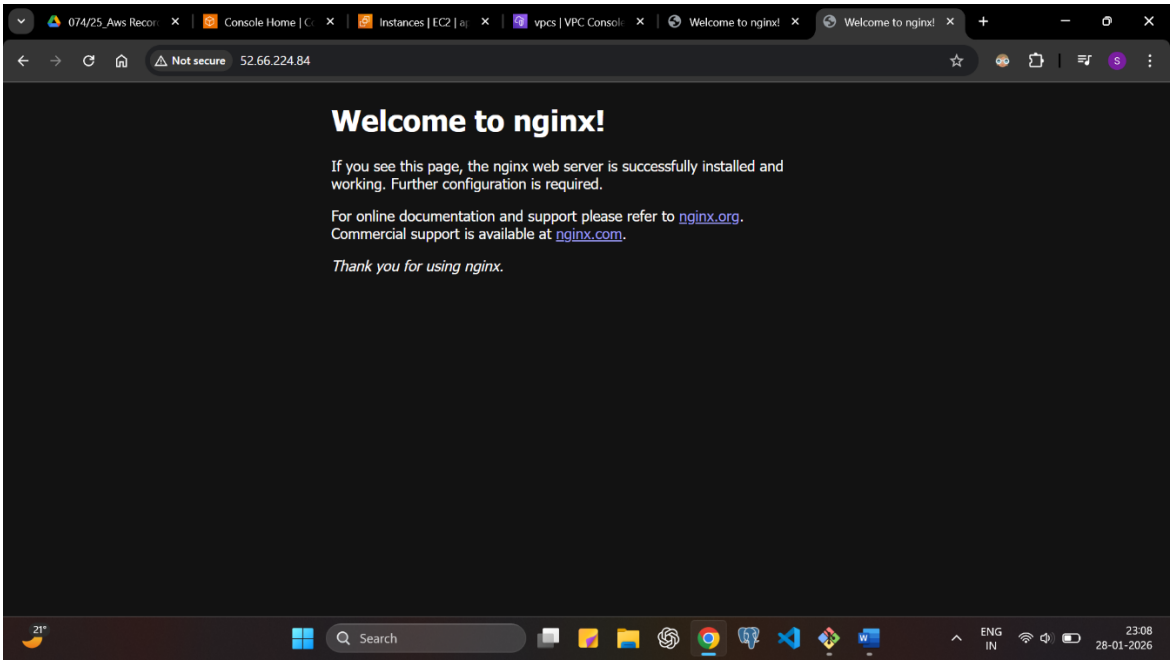
```
ubuntu@ip-10-10-3-219: ~
HP@shravani MINGW64 ~
$ cd downloads

HP@shravani MINGW64 ~/downloads
$ ssh -i "demo-key.pem" ubuntu@52.66.224.84
The authenticity of host '52.66.224.84 (52.66.224.84)' can't be established.
ED25519 key fingerprint is: SHA256:PMz1Zi07jjBQVCHX+oOKrRh8RuLMhqWsaQspZCxvWIE
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '52.66.224.84' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1018-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Wed Jan 28 17:33:55 UTC 2026

  System load:  0.26              Temperature:           -273.1 C
  Usage of /:   25.9% of 6.71GB   Processes:             115
  Memory usage: 24%               Users logged in:       0
  Swap usage:   0%                IPv4 address for ens5: 10.10.3.219

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-10-10-3-219:~$
ubuntu@ip-10-10-3-219:~$
ubuntu@ip-10-10-3-219:~$ |
```

```
root@ip-10-10-1-169:/
  Verifying        : libunwind-1.4.0-5.amzn2023.0.3.x86_64                3/7
  Verifying        : nginx-1:1.28.1-1.amzn2023.0.1.x86_64                 4/7
  Verifying        : nginx-core-1:1.28.1-1.amzn2023.0.1.x86_64            5/7
  Verifying        : nginx-filesystem-1:1.28.1-1.amzn2023.0.1.noarch      6/7
  Verifying        : nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch         7/7

Installed:
  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
  gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64
  libunwind-1.4.0-5.amzn2023.0.3.x86_64
  nginx-1:1.28.1-1.amzn2023.0.1.x86_64
  nginx-core-1:1.28.1-1.amzn2023.0.1.x86_64
  nginx-filesystem-1:1.28.1-1.amzn2023.0.1.noarch
  nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch

Complete!
[root@ip-10-10-1-169 /]#
[root@ip-10-10-1-169 /]#
[root@ip-10-10-1-169 /]# systemctl status nginx
○ nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
     Active: inactive (dead)
[root@ip-10-10-1-169 /]# systemctl start nginx
[root@ip-10-10-1-169 /]#
[root@ip-10-10-1-169 /]# systemctl start nginx
[root@ip-10-10-1-169 /]# systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
     Active: active (running) since Wed 2026-01-28 17:18:31 UTC; 43s ago
    Process: 25770 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
    Process: 25771 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
    Process: 25772 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
   Main PID: 25773 (nginx)
      Tasks: 3 (limit: 1067)
     Memory: 3.2M
        CPU: 55ms
     CGroup: /system.slice/nginx.service
             ├─25773 "nginx: master process /usr/sbin/nginx"
             ├─25774 "nginx: worker process"
             └─25775 "nginx: worker process"

Jan 28 17:18:31 ip-10-10-1-169.ap-south-1.compute.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Jan 28 17:18:31 ip-10-10-1-169.ap-south-1.compute.internal nginx[25771]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Jan 28 17:18:31 ip-10-10-1-169.ap-south-1.compute.internal nginx[25771]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Jan 28 17:18:31 ip-10-10-1-169.ap-south-1.compute.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
[root@ip-10-10-1-169 /]#
[root@ip-10-10-1-169 /]#
[root@ip-10-10-1-169 /]#
```

```
root@ip-10-10-3-219: /
amd64 Packages [29.5 kB]
Get:34 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe
Translation-en [17.9 kB]
Get:35 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe
amd64 Components [10.5 kB]
Get:36 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe
amd64 c-n-f Metadata [1444 B]
Get:37 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/restricte
d amd64 Components [216 B]
Get:38 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/restricte
d amd64 c-n-f Metadata [116 B]
Get:39 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multivers
e amd64 Components [212 B]
Get:40 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multivers
e amd64 c-n-f Metadata [116 B]
Get:41 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [141
0 kB]
Get:42 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [229
 kB]
Get:43 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [2
1.5 kB]
Get:44 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadat
a [9820 B]
Get:45 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages
[925 kB]
Get:46 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en
[209 kB]
Get:47 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Component
s [74.2 kB]
Get:48 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Met
adata [19.7 kB]
Get:49 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Package
s [2302 kB]
Get:50 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-e
n [527 kB]
Get:51 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Compone
nts [212 B]
Get:52 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Package
s [28.0 kB]
Get:53 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-e
n [6472 B]
Get:54 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Compone
nts [208 B]
Get:55 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f M
etadata [384 B]
Fetched 39.5 MB in 6s (6266 kB/s)
Reading package lists... Done
root@ip-10-10-3-219:/#
```



8. Click on public ip of ubuntu and paste it on web browser and you will see welcome window .



```
root@ip-10-10-3-219: /
Setting up nginx (1.24.0-2ubuntu7.5) ...
 * Upgrading binary nginx                                                 [ OK ]
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-10-10-3-219:/#
root@ip-10-10-3-219:/#
root@ip-10-10-3-219:/#
root@ip-10-10-3-219:/#
root@ip-10-10-3-219:/# systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-01-28 17:37:47 UTC; 1min 43s ago
       Docs: man:nginx(8)
    Process: 1723 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
    Process: 1725 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 1755 (nginx)
      Tasks: 3 (limit: 1008)
     Memory: 2.5M (peak: 5.3M)
        CPU: 30ms
     CGroup: /system.slice/nginx.service
             ├─1755 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             ├─1757 "nginx: worker process"
             └─1758 "nginx: worker process"

Jan 28 17:37:47 ip-10-10-3-219 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Jan 28 17:37:47 ip-10-10-3-219 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
root@ip-10-10-3-219:/# history
    1  cd /
    2  ls -l
    3  ln -la
    4  yum update
    5  apt-get update
    6  apt-get install nginx
    7  systemctl status nginx
    8  history
root@ip-10-10-3-219:/#
```